

AhamX Bodhi

Where I Becomes Infinite

Chain-of-Thought (CoT) Prompting

Module 2.3: Prompt Engineering | Prof. Sashikumaar Ganesan |
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Session Overview

What You Will Learn

- What chain-of-thought prompting is
- Why step-by-step reasoning improves accuracy
- Standard CoT vs. Zero-Shot CoT
- Applications in reasoning-heavy domains

Session Details

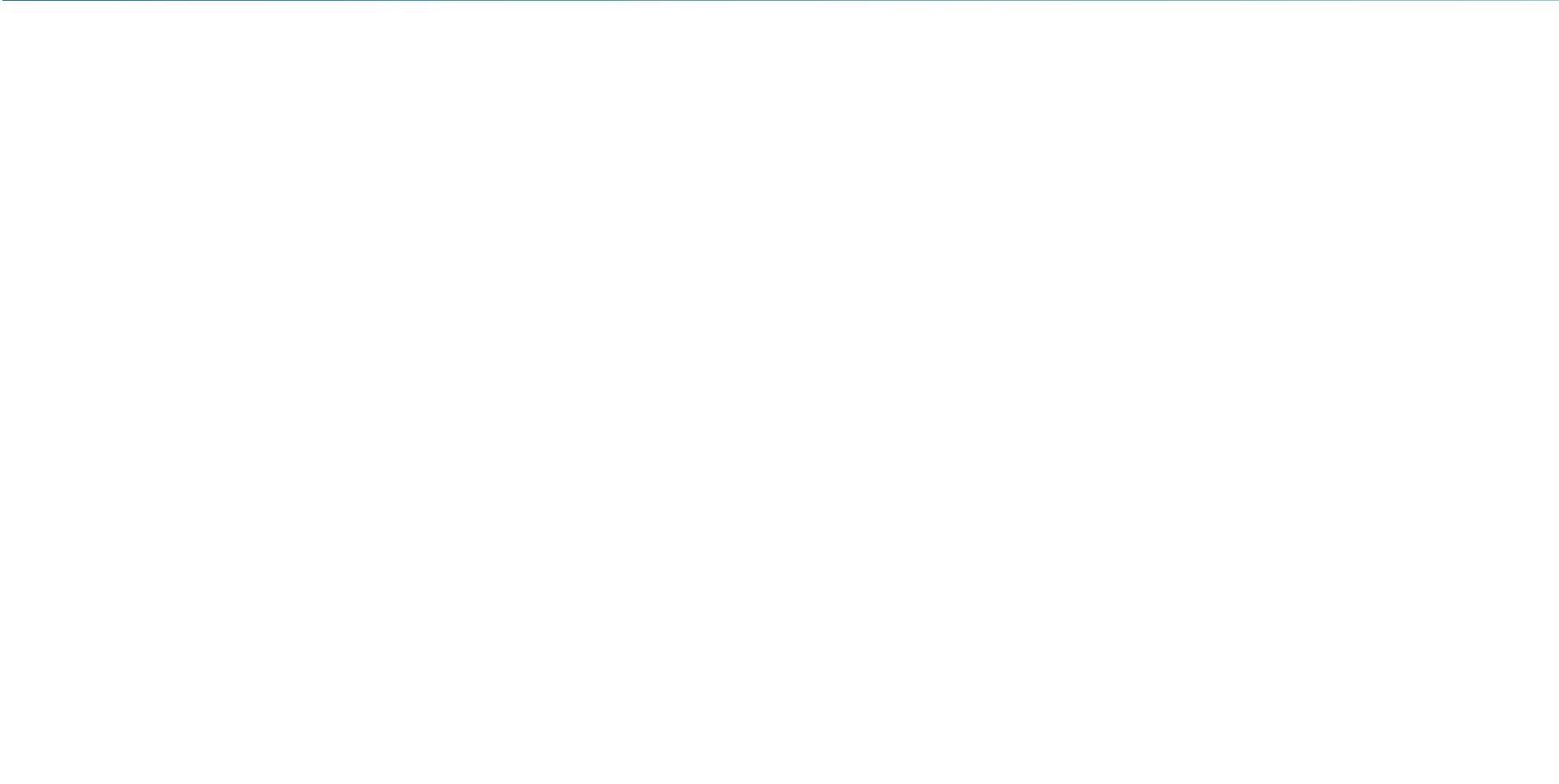
- **Duration:** 9 minutes
- **Bloom's Level:** B3 — Apply
- **Difficulty:** L2 — Intermediate
- **Module:** 2.3 Prompt Engineering
- **Prereq:** Few-Shot Prompting



Learning Objectives

By the End of This Session You Will Be Able To

- **Define** chain-of-thought prompting and its motivating research
- **Explain** why intermediate reasoning steps improve final accuracy
- **Apply** both standard CoT and zero-shot CoT ("Let's think step by step")
- **Identify** task types that benefit most from CoT reasoning





Chain-of-Thought: Definition

Core Definition

Chain-of-Thought (CoT) prompting is a technique that encourages an LLM to produce a sequence of intermediate reasoning steps before arriving at a final answer. By externalizing the reasoning process, the model makes fewer errors on complex tasks.

Research Origin

Introduced by Wei et al. (Google Brain, 2022): "Chain-of-Thought Prompting Elicits Reasoning in Large Language Models." Demonstrated dramatic accuracy gains on arithmetic, commonsense, and symbolic reasoning benchmarks in models with 100B+ parameters.



Why Chain-of-Thought Works

Cognitive Science Analogy

When humans solve a complex math problem, they write intermediate steps. LLMs benefit similarly: generating intermediate tokens forces the model to allocate computational “attention” to each reasoning substep rather than jumping directly to an answer with insufficient processing.

Technical Mechanism

- Each intermediate reasoning token becomes new context for subsequent tokens
- This creates a longer, richer computation path than direct answer generation
- Errors in intermediate steps are visible and correctable
- Works best on models $\geq 7B$ parameters; smaller models see limited benefit



Standard CoT: Few-Shot with Reasoning

Few-Shot CoT Prompt Structure

- **Instruction:** "Solve these word problems step by step."
- **Example Q:** "A store has 48 apples. They sell $\frac{1}{4}$ in the morning and 10 in the afternoon. How many remain?"
- **Example Reasoning:** "Morning sales: $48 \times \frac{1}{4} = 12$. Afternoon sales: 10. Remaining: $48 - 12 - 10 = 26$."
- **Example A:** "26 apples."
- **New Q:** "[New word problem]" → Model generates its own reasoning chain



Zero-Shot CoT: ``Let's Think Step by Step''

Simpler and Surprisingly Effective

Kojima et al. (2022) discovered that appending the phrase **“Let’s think step by step”** to a zero-shot prompt triggers chain-of-thought reasoning without requiring any examples.

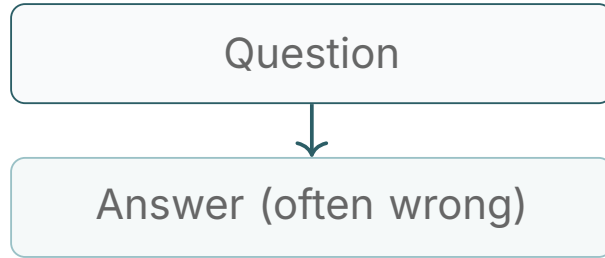
How to Apply Zero-Shot CoT

- **Step 1 prompt:** “[Problem statement] Let’s think step by step.”
- Model generates a full reasoning chain
- **Step 2 prompt:** Append reasoning + “Therefore, the answer is:”
- This two-step process (extract reasoning, then extract answer) improves accuracy
- Trigger phrases: “Let’s think step by step” / “Work through this carefully”

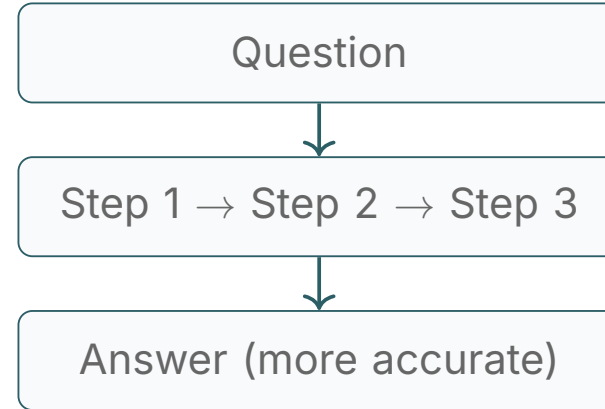


Direct Answer vs. Chain-of-Thought

Direct Answer



Chain-of-Thought



Intermediate steps decompose the problem, reducing the chance of reasoning shortcuts.



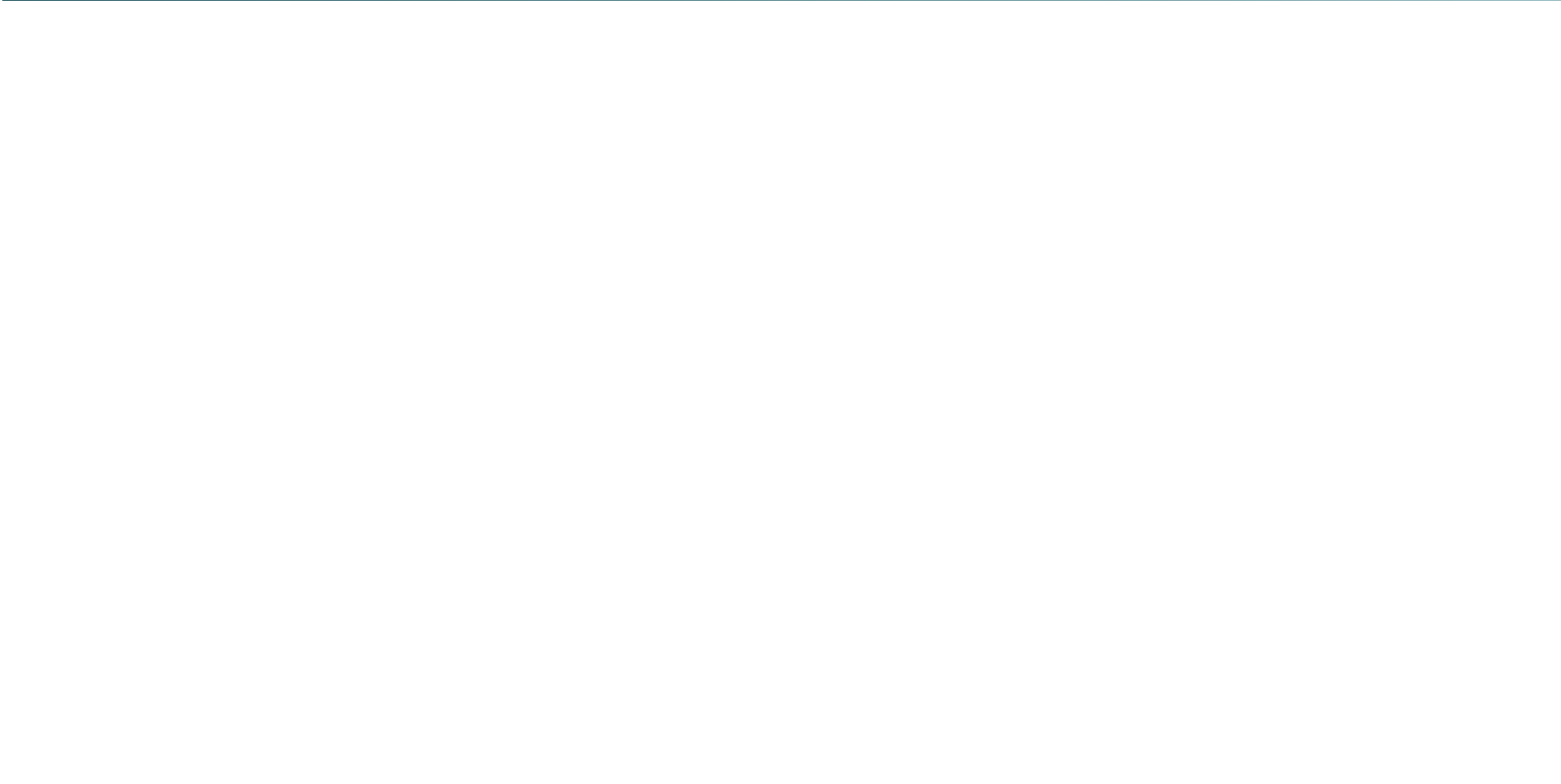
When CoT Helps Most

Strong CoT Benefit

- Arithmetic and math word problems
- Multi-step logical deductions
- Commonsense reasoning
- Code debugging (trace execution)
- Planning tasks with dependencies

Limited CoT Benefit

- Simple factual lookup tasks
- Single-step classifications
- Tasks where latency is critical
- Small models ($< 7B$ parameters)





Application: Financial Analysis and Forecasting

Multi-Step Financial Reasoning

Bloomberg AI, Morgan Stanley AI, and Goldman Sachs deploy CoT-prompted LLMs to analyze financial reports, perform multi-step ratio calculations, and generate investment thesis summaries. The reasoning chain allows auditors to verify each inference step, which is critical for regulatory compliance.

CoT in Finance

- Prompt asks model to reason through EBITDA, P/E, and DCF calculations step by step
- Each intermediate calculation is traceable and auditable
- Reduces hallucinated numbers vs. direct answer prompting



Application: Clinical Decision Support

Step-by-Step Diagnostic Reasoning

Systems like Meditron (EPFL), Med-PaLM 2 (Google), and clinical assistants built on GPT-4 use CoT prompting to guide diagnosis. The model is prompted to reason through symptoms, differential diagnoses, and test results before suggesting a diagnosis.

Why CoT Matters in Healthcare

- Clinicians can review each reasoning step, not just the final answer
- Improves performance on USMLE-style medical questions by 10–20%
- Reduces direct hallucination of diagnoses without supporting evidence



Application: Code Debugging via CoT

Trace-Based Debugging

GitHub Copilot Chat, Cursor, and JetBrains AI Assistant use CoT-style prompts to debug code by first tracing execution, then identifying the error, then suggesting a fix — mirroring how human developers debug systematically.

CoT Debugging Prompt Pattern

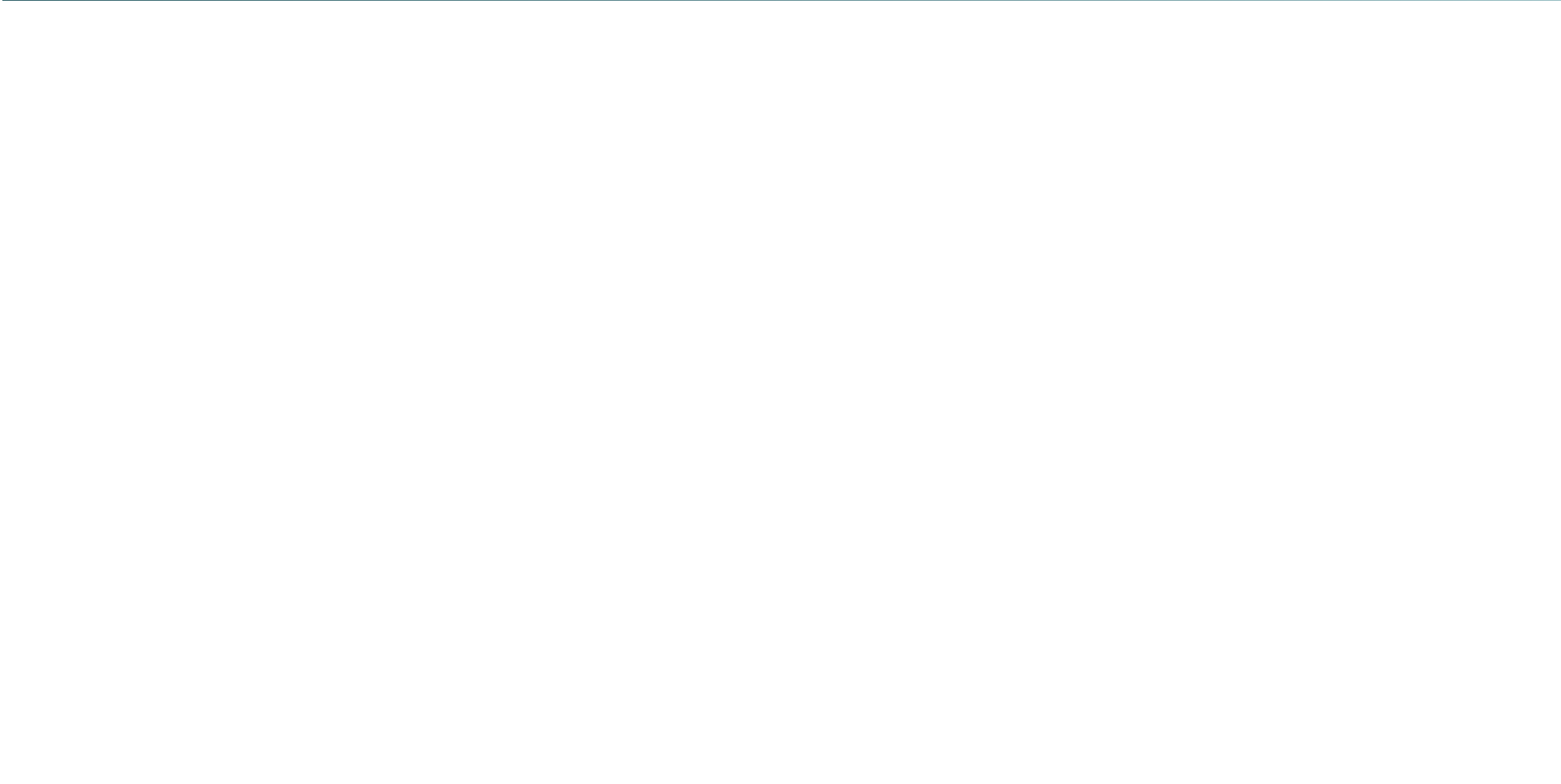
- "Trace through this function step by step with input X."
- Model narrates execution: "Line 3: variable a = 5. Line 5: division by zero."
- Then proposes fix based on identified root cause



CoT Variants and Extensions

Beyond Standard CoT

- **Self-Consistency CoT** (Wang et al., 2022): Generate multiple reasoning chains, take majority vote on final answer. Improves accuracy further.
- **Tree-of-Thought (ToT)** (Yao et al., 2023): Explore multiple reasoning branches in parallel; backtrack on dead ends. Better for open-ended planning.
- **Program-of-Thought**: Model writes Python code as the reasoning chain, then executes it. Precise for arithmetic.
- **ReAct** (Yao et al., 2022): Interleaves reasoning with tool use (search, code execution).





Key Takeaways

Core Concepts to Remember

- CoT prompting elicits step-by-step reasoning before the final answer
- Standard CoT uses few-shot examples with reasoning chains demonstrated
- Zero-Shot CoT uses “Let’s think step by step” without examples
- CoT improves accuracy on arithmetic, logic, commonsense, and code tasks
- Variants include Self-Consistency, Tree-of-Thought, and ReAct

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